

Problem 53. Let n, m be positive integers such that $n \geq 2m + 1$. Given n distinct points and nm distinct line segments, each joining a pair of points, prove that there exist distinct points $A_0, A_1, \dots, A_m$, so that for each $i = 0, 1, \dots, m - 1$, there is a line segment joining A_i to A_{i+1} .

Problem 54. Find all functions $f: \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f(-f(x) - f(y)) = 1 - x - y$ for all $x, y \in \mathbb{Z}$.

Problem 55. Let $n \geq 3$ be an odd integer. Suppose you play the following game using an $n \times n$ grid made up of n^2 unit squares. The game consists of n^2 turns. At each turn the following actions must be performed in the order specified:

- Choose an empty square and insert a positive integer greater than or equal to 1 and less than or equal to n^2 . Throughout the game, integers can be used only once.
- You gain 1 point if the sum of the integers written in the squares lying on the same row as the square you chose is a multiple of n . The same rule applies for columns. You get 2 points if both the row sum and the column sum are multiples of n .

Find the maximum possible value for the total sum of the points you gain by the end of the game.

Problem 56. In $\triangle ABC$, let I be its incentre, ω be its incircle, M be the midpoint of BC , and K be the point of intersection of the line passing through A and perpendicular to line BC , and the line passing through M and perpendicular to the line AI .

Prove that the circle with AK as diameter is tangent to ω .